American Computer Science League

2023-2024 ● Contest 2: Short Problems ● Junior Division

1. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

− + 2 * 3 8 / * 6 2 3

A. 12
B. 20
C. 22
D. 25
E. 30

2. Prefix/Infix/Postfix Notation

Convert the following infix expression to postfix:

$$\frac{A+B}{C-1} + \frac{2A-3}{D}$$

A. AB+C1−/2A3*−D/+
B. AB+C1−/2A*3−D/+
C. ABC1−+/2A*3−D/+
D. AB+C1−/2A*3−D+/
E. AB+C1−/2A*3D−/+

3. Bit-String Flicking

Evaluate:

00110 OR 01100 XOR 11010

A. 01001
B. 11001
C. 00110
D. 01010
E. 10110

4. Bit-String Flicking

Evaluate:

(LSHIFT-1 01110) AND (RCIRC-2 (10110 OR NOT 10101))

A. 10100
B. 11111
C. 11100
D. 00011
E. 01001

5. What Does This Program Do? (Looping)

What is the following program's output?

```
p = .01
d = 1
t = .01
while t < d
     d = d + 1
     p = p * 2
     t = t + p
end while
output d
```

A. 7
B. 8
C. 9
D. 10
E. 11

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1. 前缀/中缀/后缀

如果下方所有数字都是一位数，请计算下方前缀表达式：

− + 2 * 3 8 / * 6 2 3

A. 12
B. 20
C. 22
D. 25
E. 30

2. 前缀/中缀/后缀

请将下方中缀表达式转换为后缀表达式：

$$\frac{A+B}{C-1} + \frac{2A-3}{D}$$

A. AB+C1−/2A3*−D/+
B. AB+C1−/2A*3−D/+
C. ABC1−+/2A*3−D/+
D. AB+C1−/2A*3−D+/
E. AB+C1−/2A*3D−/+

3. 比特位操作

请计算下列表达式：

00110 OR 01100 XOR 11010

A. 01001
B. 11001
C. 00110
D. 01010
E. 10110

4. 比特位操作

请计算下列表达式：

(LSHIFT-1 01110) AND (RCIRC-2 (10110 OR NOT 10101))

A. 10100
B. 11111
C. 11100
D. 00011
E. 01001

5. 这个程序是做什么的？（循环）

请问以下程序的输出值是多少？

```
p = .01
d = 1
t = .01
while t < d
     d = d + 1
     p = p * 2
     t = t + p
end while
output d
```

A. 7
B. 8
C. 9
D. 10
E. 11